

Part (2) - Data Analysis:

Tableau Calculated Fields / Metric Definitions:

Order Match Time Seconds

Definition: The time difference in seconds between when the order was created and when the driver responded.

Formula: DATEDIFF('second', [Order Create Timestamp], [Driver Response Timestamp])

Order Match Time Minutes

Definition: Order match time converted from seconds to minutes.

Formula: [Order Match Time Seconds] / 60

Experiment Group

Definition: Orders are split into pre-change and post-change groups based on the system change timestamp.

Formula:

IF [Order Create Timestamp] < #2017-03-30 00:00:00# THEN "Pre"

ELSE "Post"

END

Order Date

Definition: The date when the order was created.

Formula: DATE([Order Create Timestamp])

Order Hour

Definition: The hour of day when the order was created.

Formula: DATEPART('hour', [Order Create Timestamp])

Long Match > 60s

Definition: Indicator for whether an order took more than 60 seconds to be matched.

Formula: IF [Order Match Time Seconds] > 60 THEN 1 ELSE 0 END

Long Match > 180s

Definition: Indicator for whether an order took more than 180 seconds to be matched.

Formula: IF [Order Match Time Seconds] > 180 THEN 1 ELSE 0 END

P25 Match Time

Definition: The 25th percentile of order match time. 25% of orders were matched within this time.

Formula: PERCENTILE([Order Match Time Seconds], 0.25)

P50 Match Time

Definition: The 50th percentile of order match time, also known as the median.

Formula: PERCENTILE([Order Match Time Seconds], 0.50)

P75 Match Time

Definition: The 75th percentile of order match time. 75% of orders were matched within this time.

Formula: PERCENTILE([Order Match Time Seconds], 0.75)

P90 Match Time

Definition: The 90th percentile of order match time. 90% of orders were matched within this time, and the remaining 10% took longer.

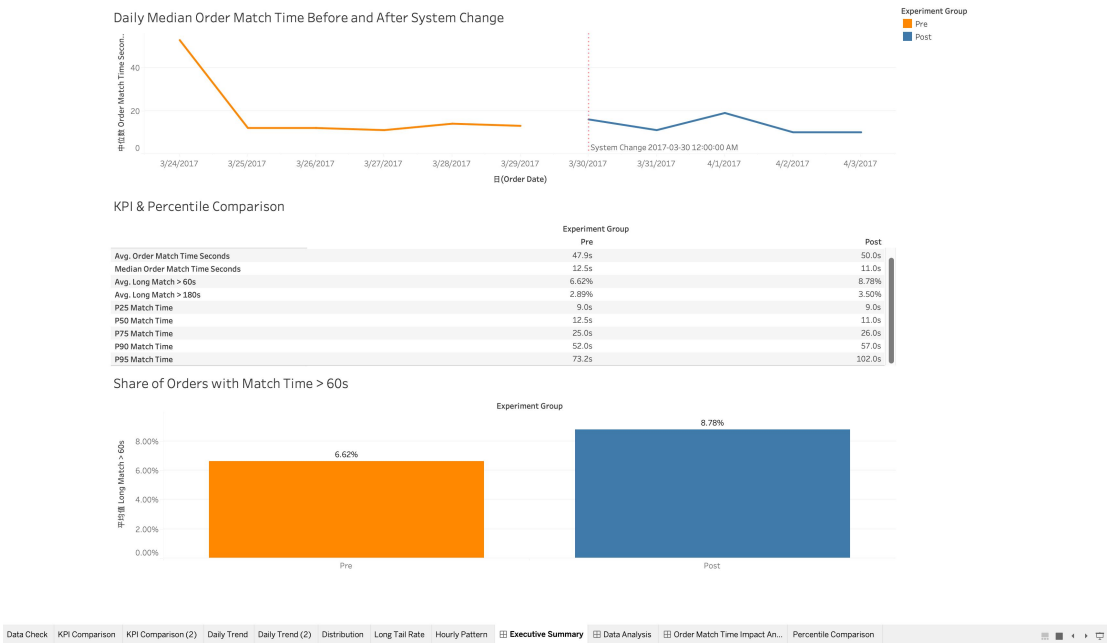
Formula: PERCENTILE([Order Match Time Seconds], 0.90)

P95 Match Time

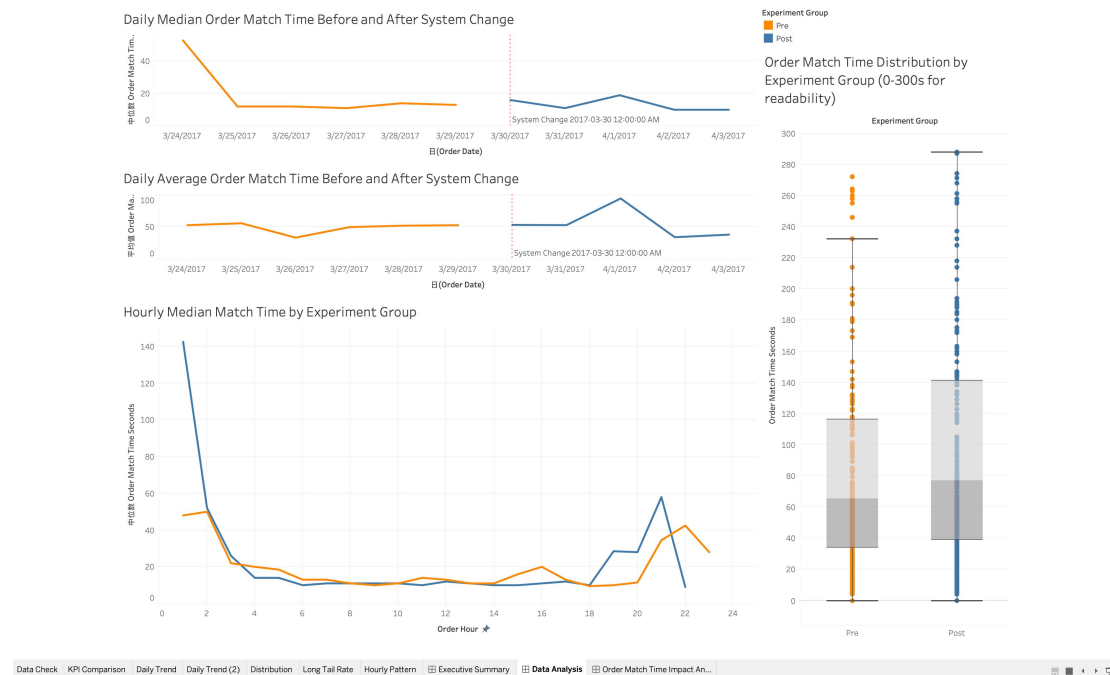
Definition: The 95th percentile of order match time. 95% of orders were matched within this time, and the remaining 5% took longer.

Formula: PERCENTILE([Order Match Time Seconds], 0.95)

Dashboard 1 — Executive Summary



Dashboard 2 — Data Analysis



a) What is the resulting impact of this change?

The system change does not show a clear overall improvement in Order Match Time. After the change, median match time decreased slightly from 12.5 seconds to 11.0 seconds, suggesting that typical orders may have been matched faster. However, average match time increased from 47.9 seconds to 50.0 seconds, and the share of orders taking more than 60 seconds increased from 6.62% to 8.78%, and the share of orders taking more than 180 seconds increased from 2.89% to 3.50%.

This suggests that the change may have improved normal cases but did not improve long-tail slow-match orders. Overall, I would not conclude that the change clearly improved order allocation performance.

b) Are there any additional insights or observations that you think are worth noting?

Order Match Time is highly right-skewed. Most orders were matched quickly, but a small number of very slow orders pulled up the average. Therefore, median and long-tail metrics such as P90, P95, and the >60s rate are more informative than average alone.

The daily trend also does not show a strong and consistent post-change improvement. Although the overall median improved slightly, some post-change days still had higher median match time. The hourly pattern further suggests that match time varies by time of day, so the impact of the system change may not be consistent across all hours.

The percentile comparison shows that P90 and P95 worsened after the change, suggesting that the change did not improve slow-match cases.